

Logic Gates

What is a gate?

→ The building blocks used to create digital circuits are logic gates.

→ Combination of transistors that performs binary logic

→ The function of each gate can be represented by a truth table or using Boolean notation.

Types of Logic Gates

→ Basic gates

NOT, OR, AND, BUFFER

Derived Gates

NOR

NAND

→ Universal Gates

NOR

NAND

→ Specific Gates

EX-OR

EX-NOR

NOT Gate

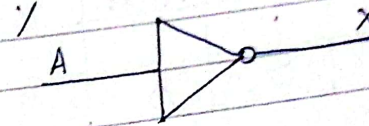
→ A NOT gate accepts one input value and produces one output value.

→ A NOT gate is sometimes referred to as an **inverter** because it inverts the input value.

Boolean Expression:

$$X = A'$$

Logic Diagram Symbol:



Truth table:

A	$X = A'$
0	1
1	0

BUFFER Gate:

Expression:

$$X = A$$

Logic Diagram:



Truth table:

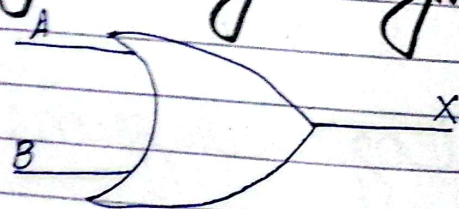
A	X=A
1	1
0	0

OR-Gate

Boolean Expression:

$$X = A + B$$

Logic Diagram Symbol:



Truth table:

for two input :

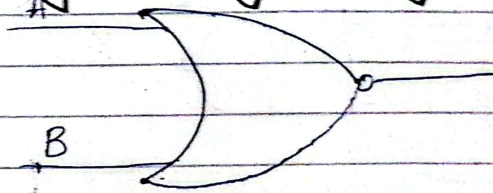
NOR Gate

⇒ Opposite of OR gate.

Boolean Expression:

$$X = (A + B)' = (\overline{A + B})$$

Logic Diagram Symbol:



Truth table:

A	B	A+B	X=(A+B)'
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

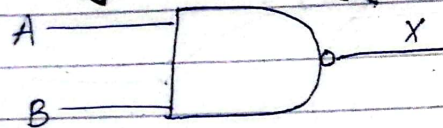
NAND Gate

→ Opposite of AND gate.

Boolean Expression:

$$X = (A \cdot B)' = (\overline{A \cdot B})$$

Logic Diagram Symbol:



Truth table:-

A	B	$A \cdot B$	$\overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

A	B	$X = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

for three input values:

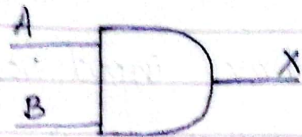
A	B	C	$X = A + B + C$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

AND Gate

→ An AND gate accepts two input signals.

Boolean Expression:
 $X = A \cdot B$

Logic Diagram Symbol:



Truth table:

A	B	$X = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

A	B	C	$X = A \cdot B \cdot C$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Prove that $A+B = \overline{A \cdot B}$

A	B	$\overline{A}$	$\overline{B}$	$A+B$	$\overline{A \cdot B}$
0	0	1	1	0	1
0	1	1	0	1	1
1	0	0	1	1	1
1	1	0	0	1	0

Hence, $A+B = \overline{A \cdot B}$

Prove that $\overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$

A	B	C	$\overline{A+B+C}$	$\overline{A} \cdot \overline{B} \cdot \overline{C}$
0	0	0	1	1
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

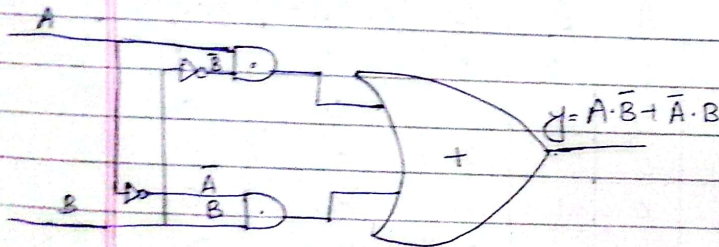
Hence $\overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$

Boolean

Prove that $\overline{A \cdot B} = \overline{A} + \overline{B}$

A	B	$\overline{A}$	$\overline{B}$	$\overline{A+B}$	$\overline{A \cdot B}$
0	0	1	1	1	1
0	1	1	0	1	1
1	0	0	1	1	1
1	1	0	0	0	0

Exclusive-OR gate:



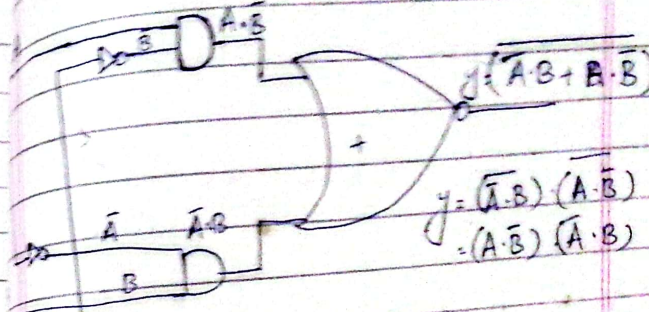
Logical Expression:

$Y = A \cdot \overline{B} + \overline{A} \cdot B$

Truth Table:

A	B	$\overline{A}$	$\overline{B}$	$\overline{A \cdot B}$	$A \cdot \overline{B}$	$\overline{A} \cdot B$	$Y = \overline{A \cdot B} + A \cdot \overline{B} + \overline{A} \cdot B$
0	0	1	1	0	0	0	0
0	1	1	0	1	0	1	1
1	0	0	1	1	1	0	1
1	1	0	0	0	0	0	0

Exclusive-NOR Gate:

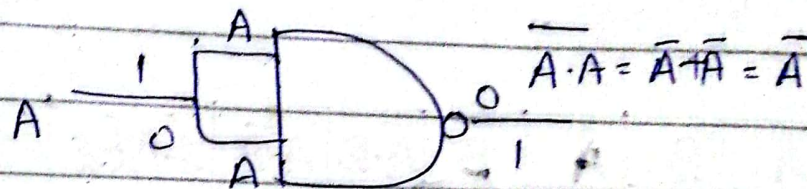
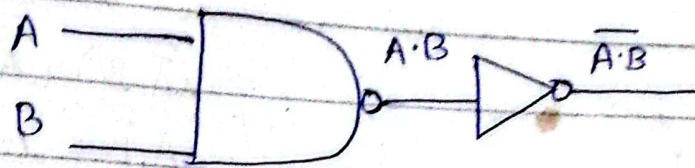
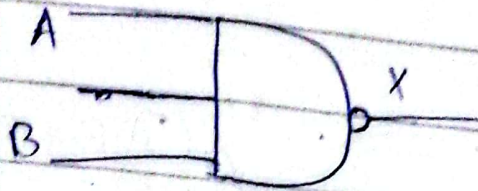


A	B	$\overline{A}$	$\overline{B}$	$\overline{A \cdot B}$	$A \cdot \overline{B}$	$\overline{A} \cdot B$	$Y = \overline{A \cdot B} + A \cdot \overline{B} + \overline{A} \cdot B$	Y'
0	0	1	1	0	0	0	0	1
0	1	1	0	1	0	1	1	0
1	0	0	1	1	1	0	1	0
1	1	0	0	0	0	0	0	1

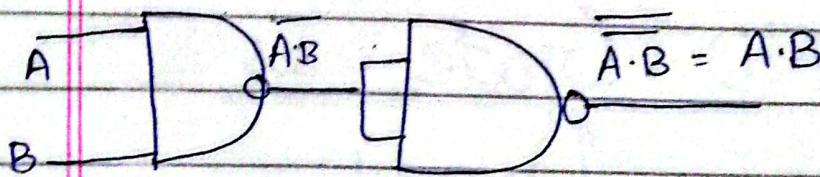
NAND Gate as Universal gate:

→ A gate that can be converted into other fundamental gates (NOR, OR, AND) is called universal NAND gate.

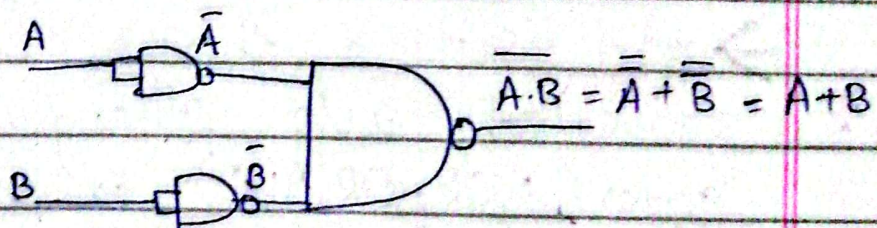
NOT:



AND:



OR:



EX NOR

